

STERLITE COPPER: PROFIT VERSUS PUBLIC HEALTH¹

Veena Keshav Pailwar wrote this case solely to provide material for class discussion. The author does not intend to illustrate either effective or ineffective handling of a managerial situation. The author may have disguised certain names and other identifying information to protect confidentiality.

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On May 28, 2018, considering the violent protests against environmental pollution taking place, the Government of Tamil Nadu, a state in southern India, ordered a permanent shutdown of Sterlite Copper (Sterlite), a copper smelting plant located in Tuticorin (also known as Thoothukudi), a port city in the state.² The protestors opposed the proposed expansion of the copper plant on the grounds that the existing plant was polluting the air and the groundwater, which had led to a surge in rates of cancer and respiratory diseases in the neighbouring community. Despite denial from district officials of any link between the operations of the plant and the health of the local population, the state government had passed an order for the closure of the plant because the protests had gotten out of hand. With Sterlite's significant contribution to Vedanta Limited (Vedanta), its holding company, the share price of Vedanta fell to a 10-month low in reaction to the shutdown order.³ Speaking about the loss to the company due to the shutdown, P Ramnath, chief executive officer of Sterlite, said, "A total (annual) turnover of ₹600 billion⁴ is affected."⁵ Denying the charges of pollution against the plant, Ramnath clarified that the major pollutants in the region were several thermal power plants, which were responsible for 99 per cent of the sulphur dioxide in the region.⁶ Refuting the pollution charges, he stated, "False propaganda dominated over the truth about the company. . . we will continue to bring out the truth to the general public and explore all options."⁷ The business environment was hostile as a result of the situation, but the closure of the plant would result in a huge loss for Vedanta. What should Vedanta do? Should it take legal action or should it close down the plant?

COPPER INDUSTRY IN INDIA

Copper was the third most widely-used metal globally, after only steel and aluminum. It was used in electricity to telecommunications, railway transport, defense, and astronautics products. As copper was a good conductor of electricity, the electricity sector was the largest user of copper in India (see Exhibit 1). The country's total consumption of refined copper was around 650,000 tonnes.⁸ This usage was expected to rise to 1.433 million tonnes by 2026⁹ due to growing demand from proposed infrastructure projects, renewable energy projects, rural and railway electrification projects, and expanding building activities.¹⁰ The focus on copper-intensive electric vehicles in the country was expected to boost demand further, to around 2.5 million tonnes by 2026.¹¹

Copper metal was extracted by the smelters from copper ore (which was primarily sulfide copper mineral containing 0.5 to 2 per cent of copper by mass) or copper concentrate (which was extracted from copper ore and contained about 25 per cent copper by mass).¹² With 813 million tonnes, the state of Rajasthan (see

Exhibit 2) had the largest reserves of copper ore in the country—53.81 per cent of the total reserves in the country.¹³ Jharkhand followed with 295 million tonnes (19.54 per cent) and then Madhya Pradesh with 283 million tonnes (18.75 per cent).¹⁴ The copper reserves in the country constituted just 2 per cent of the world reserves.¹⁵ Copper ore and copper concentrate together contributed less than 1 per cent to both the total volume of production and total employment in the mining and quarrying sector.¹⁶ The mining and quarrying sector itself accounted for just 2.5 per cent of the gross value added of India.¹⁷ With an employment elasticity of 0.52 per cent, the mining sector had 13 times and six times more potential to create employment than the agriculture and manufacturing sectors, respectively, for every 1 per cent of growth in gross domestic product.¹⁸ Despite abundant mineral reserves, the country was not able to realize its potential in this sector, owing to strict regulations on mines and mineral exploration.¹⁹

The three major players in the Indian copper industry were Hindalco Industries Limited, Vedanta (or Sterlite), and Hindustan Copper Limited (see Exhibits 3 and 4). Hindustan Copper Limited was a public sector company involved in the mining of ore as well as in the smelting, refining, and casting of refined copper. Hindalco Industries Limited and Vedanta were private sector companies that owned mines in other countries. As the copper ore reserves in India were limited, these companies had to meet almost 100 per cent of copper concentrate requirements for their smelters from imported copper. India imported copper ore from Chile, Indonesia, Australia, and Canada.²⁰ Although India imported copper ore, its copper smelters had enough capacity to meet almost 98 per cent of the domestic requirement for refined copper.²¹ The country also exported a small amount of refined copper to other countries. However, India's consumption was rapidly catching up with its production, necessitating expansion in production capacity.²²

STERLITE: HISTORY AND ECONOMICS

Vedanta Limited, a part of Vedanta Resources Limited—one of the world's largest mining and metal conglomerates—was permitted by the Tamil Nadu Pollution Control Board (TNPCB) in 1994 to operate a copper project. It received environmental clearance from the Ministry of Environment and Forests (MOEF) in 1995, and a licence to operate from the TNPCB in 1996.²³ The Sterlite plant was set up by Vedanta in Tuticorin, a densely populated town with a population of 527,000 in 2018,²⁴ because of the town's proximity to the port and investor-friendly environment.²⁵ The plant was located in the State Industries Promotion Corporation of Tamil Nadu complex, which was an industrial belt spread over an area of around 1,500 acres housing nearly 80 industries,²⁶ and producing chemicals, marine products, salt, spices, cold storage, high-density polyethylene food bags, and textiles.²⁷

Sterlite had an annual capacity of around 400,000 tonnes of refined copper (copper anodes)²⁸ (see Exhibit 4). Of its total production, the plant supplied around 55 per cent to the domestic market and exported the rest, mainly to China.²⁹ The plant met almost one-third of the copper demand in India.³⁰

Sterlite catered to over 400 small and medium enterprises in the downstream value chain (see Exhibit 5).³¹ Its main products were copper cathode and copper rod. Apart from supplying these products to several industries, which mainly consisted of electrical, telecommunications, transport, and engineering companies, Sterlite also catered to several chemical, fertilizer, and detergent companies by supplying sulphuric acid and di-ammonia phosphate. The other by-products it supplied in the market were copper slag (mainly used in the building and road construction industry) and anode slime, which was used in precious metal refinery to extract gold and silver (see Exhibit 5).

With annual sales of ₹220 billion,³² Sterlite contributed 5 per cent to the operating profits (earnings before interest, tax, depreciation, and amortization) of Vedanta³³ (see Exhibit 6), and 3 per cent (₹20 billion) to Tamil Nadu's gross domestic product.³⁴ By reducing the need for imports, the plant could also save foreign currency worth around ₹140 billion.³⁵

In terms of turnover, Sterlite was the fifth largest company in Tamil Nadu and supported around 4,000 employees directly.³⁶ About 70 per cent of these employees were from Tamil Nadu; of these, 70 per cent were from Tuticorin.³⁷ The company also supported around 20,000 employees, indirectly, in the small and medium-sized enterprises in downstream (customer) industries,³⁸ and benefited nearly 200,000 people under its corporate social responsibility activities.³⁹

Vedanta had invested ₹25 billion in the establishment of Sterlite.⁴⁰ Taking account of the growing demand for copper, the company wanted to expand its capacity from 0.4 million tonnes a year to 0.8 million tonnes a year by setting up another unit in the State Industries Promotion Corporation of Tamil Nadu complex.⁴¹ For the proposed expansion, the company had received environmental clearance in 2009 and the permission to operate in 2016.⁴² The company had already committed ₹20 billion of the ₹35 billion earmarked for the expansion⁴³ and had also started hiring people for the proposed expansion.⁴⁴

PUBLIC CONCERNS AND PROTESTS

Sterlite had been the subject of several controversies in the past. Before moving to Tamil Nadu, Vedanta had received permission to open a copper plant in the Ratnagiri district of Maharashtra. However, due to public agitation, the Government of Maharashtra withdrew the permission in 1993.⁴⁵ In addition, two other states, Gujarat and Goa, had also rejected Vedanta's request to open a plant in those states.⁴⁶

Within a year of the cancellation of permission to set up a plant in Maharashtra, the company obtained permission from the TNPCB to open a plant in Tamil Nadu. The TNPCB granted the permission on the condition that the factory be located at least 25 kilometres (km) away from the Gulf of Mannar.⁴⁷ The plant also received environmental clearance from the MOEF with the same condition.⁴⁸ However, the Sterlite plant ultimately was built within 14 km of the Gulf of Mannar,⁴⁹ amounting to a violation of the aforementioned condition. Since Sterlite was a zero-discharge company, the Government of Tamil Nadu ignored the location violation clause and in October 1996 gave it a licence to operate.⁵⁰ While granting the licence, the government stipulated that the company develop a green belt within a 25-metre radius of the plant and that its functioning not contaminate the groundwater or the air around it.⁵¹

Protests against Sterlite in Tamil Nadu started soon after the proposed plant received the environmental clearances from the TNPCB and the MOEF. In 1996, the National Trust for Clean Environment challenged the permission in the Madras High Court.⁵² In November 1998, on the directive of the High Court, the National Environmental Engineering Research Institute (NEERI) submitted a study on the plant.⁵³ The study found that the plant contaminated the groundwater with arsenic, lead, and selenium, along with other chemicals, and that the gas that leaked from the plant was making the nearby workers sick. Based on the report, the Madras High Court ordered the immediate closure of the plant.⁵⁴ However, the company succeeded in getting a stay order. Later, the court modified its earlier order and instructed the NEERI to conduct another study.⁵⁵ This study, along with many other studies carried out by the institute, defended Sterlite's operations. The case was prolonged for more than a decade. In September 2010, citing a violation of the law and the pollution norms, the Madras High Court ordered Sterlite Copper to shut down its operations,⁵⁶ but the Supreme Court stayed the order. In April 2013, it overturned the High Court directive stating,

“The plant of the appellants contributes substantially to the copper production in India and copper is used in defence, electricity, automobile[s], construction and infrastructure, etc. The plant of the appellants has about 1,300 employees and it also provides employment to large number of people through contractors . . .⁵⁷

The Supreme Court overturned the shutdown order of the High Court but imposed a fine of ₹1 billion on the company for violating the environmental norms.⁵⁸

Much before the Madras High Court order and the final verdict of the Supreme Court, in January 1997, Sterlite commenced its operations.⁵⁹ However, within a few months of the commencement of operations, workers in the nearby offices and establishments started complaining of respiratory problems. Several cases were filed against the company with the TNPCB. But each time a case was filed, Sterlite managed to come out exonerated. In the subsequent period, in March 2013, residents of Tuticorin complained of coughing, wheezing, eye irritation, and miscarriages.⁶⁰ They attributed these problems to a gas leak from Sterlite and demanded it be shut down. The TNPCB ordered the shutdown of the plant.⁶¹ However, within three months, the National Green Tribunal (see Exhibit 7) directed the reopening of the plant on the grounds that there was no scientific data supporting the emission in excess of the prescribed parameters.⁶²

Another round of protests against Sterlite began in February 2018, when the company expressed its intention to double production by expanding plant capacity.⁶³ The protests, which had been initiated by 250 people, gradually grew in scale as an increasing number of people from Tuticorin and the neighbouring areas joined.⁶⁴ Amid the series of protests, citing annual scheduled maintenance as the reason, the plant was shut down in March 2018.⁶⁵ Meanwhile, Vedanta appealed to the TNPCB to seek consent to reopen the plant. The TNPCB rejected the plea on the grounds that the company had violated the environmental norms. With many national and international non-governmental organizations and many political parties supporting the anti-Sterlite movement, and with the movement becoming increasingly stronger, the pressure to keep the plant closed was sustained. The demonstrations, which had been peaceful for 100 days, took a violent turn on May 22, when the police tried to disperse the protestors by firing at the crowd. The police firing killed at least 11 people and injured several protestors.⁶⁶ Considering the violence, the Government of Tamil Nadu was compelled to issue an order for the plant's permanent shutdown. The non-governmental organizations; human rights activists; and Tamil solidarity groups involved in mobilizing the masses against Vedanta, such as Foil Vedanta, the Tamilians in the United Kingdom, the Periyar Ambedkar Study Circle, the South Asia Solidarity Group, Tamil Solidarity, Parai Voice of Freedom, and Veera Tamilar Munnani, also gathered the support of the Labour Party, the main opposition party in the United Kingdom.⁶⁷ Considering Vedanta a rogue company, the Labour Party demanded a delisting of its stocks from the London Stock Exchange to prevent damage to the reputation of London's financial markets.⁶⁸ The solidarity groups also mustered the support of the Tamil diaspora in the United States against the killing in Tuticorin.⁶⁹

Vedanta's other enterprises had also faced problems in other states, including Chhattisgarh, Odisha, and Rajasthan.⁷⁰ In Chhattisgarh, Vedanta had been accused of malpractice. In 2001, Vedanta was awarded alumina refinery, smelter, and bauxite mines owned by Bharat Aluminium Company Limited at well below their true worth.⁷¹ In September 2009, several workers died under the aluminum smelter due to the collapse of a chimney under construction.⁷² The subsequent inquiry found Vedanta guilty of the incident. However, Vedanta was able to have the inquiry report suppressed.⁷³ In August 2013, tribes in Odisha's Niyamgiri Hills rejected Vedanta's bauxite mining project because they feared that through it they would lose their identity; their good health; their livelihood; and their expert knowledge of the hills, farming, and forests.⁷⁴ As a result, Vedanta was forced to cancel its bauxite mining project in Niyamgiri Hills but continued to operate its refinery located in Lanjigarh, a place close to the Hills.⁷⁵ Because of high pollution caused by the refinery, local communities demanded that even the refinery be decommissioned.⁷⁶ Despite the protest over its operations, Vedanta continued to operate the refinery using imported bauxite from other states and abroad. Finally, considering developmental needs, the Government of Odisha awarded two mines to Vedanta from other parts of the state.⁷⁷ In 2014, a public interest litigation was filed against the sale of a majority stake in Hindustan Zinc Limited, a public sector company located in Rajasthan, by the Government of India, at a throwaway price to Vedanta in 2002–2003.⁷⁸ However, the Supreme Court dismissed the petition, declaring it frivolous.⁷⁹ Vedanta Resources Limited, the holding company of Vedanta, had also been facing protests in other countries, such as Zambia, for environmental pollution.⁸⁰ In Zambia, Vedanta was charged in 2004 for buying subsidiary Konkola Copper Mines (KCM) at a much undervalued price.⁸¹

Subsequent investigations by the Zambian government indicated that Vedanta was hiding its profits by transfer pricing, i.e., selling copper at highly underpriced rates to its own subsidiary, Fujairah Gold, in Dubai.⁸² In another incident, in 2011, the High Court of Zambia imposed a heavy penalty on Konkola Copper Mines for the release of pure acid and toxic mine waste into the Kafue River over several days, which affected the health of the local community.⁸³ However, the Supreme Court of Zambia reduced the penalty substantially.⁸⁴

STERLITE'S ENVIRONMENTAL SUSTAINABILITY MEASURES

Sterlite had invested more than ₹5 billion in best-class technologies to adopt various environmental protection measures, such as flue-gas desulphurization units with bag filters, modern technology-based reverse osmosis plants, and evaporators.⁸⁵ Sterlite was the fourth-lowest energy consumption copper smelter in the world. For several years, the company had been reporting greenhouse gas emissions through the Carbon Disclosure Project, had been a zero-liquid discharge plant since its inception, and disposed of its waste material through its secured landfill facility. In order to control environmental pollution, it had built a green cover in and around the periphery of the plant covering an area of 68 hectares.⁸⁶ In order to ensure social involvement, acceptability, and sustainability, Sterlite had been focusing on education, health, childcare, women empowerment, infrastructure, livelihood, agriculture and animal husbandry, and employee voluntarism under its corporate social responsibility activities. Due to its sustainability measures, it was accredited with a “five star” rating by the British Safety Council.⁸⁷

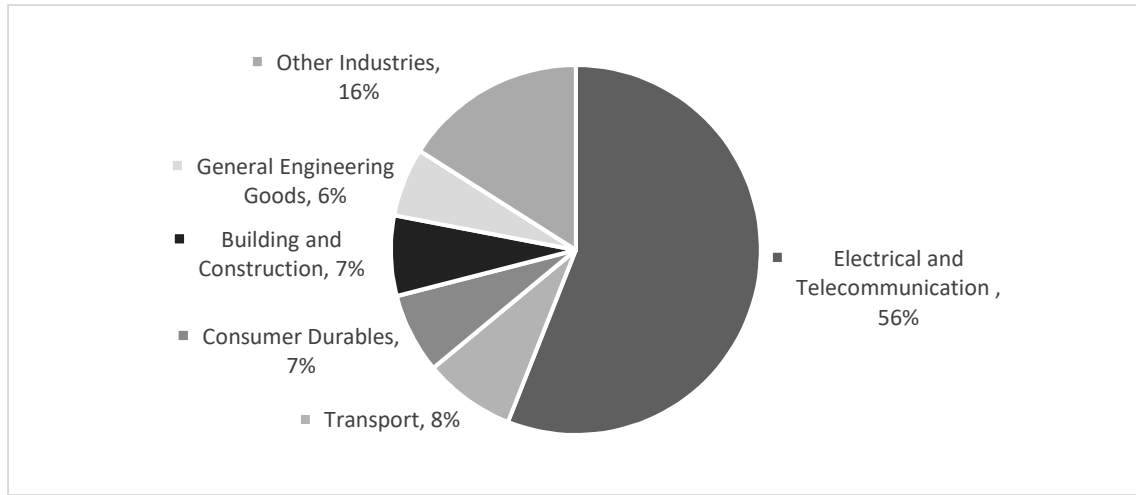
WHAT THE DATA SAID

Apart from the studies conducted by the NEERI, several other studies were carried out by the government and by non-government bodies to assess the environmental pollution and health status of the people living near the plant. Though most of the studies found severe water and air pollution, they failed to attribute this to Sterlite. For example, a study commissioned by the government and carried out by the Department of Community Medicine at Tirunelveli Medical College in 2006–2007 to analyze the health status of the people living within a 5-km radius of the plant, found a high prevalence of respiratory-tract infections in the study area. The infections could be attributed to the air pollution arising from the release of the gases and the particulate matter.⁸⁸ However, the study could not strongly attribute the pollution to Sterlite.⁸⁹ Besides this, it could not find arsenic traces in the groundwater, and none of the skin-related diseases detected during the study period were from arsenic poisoning.⁹⁰

WHAT SHOULD VEDANTA DO?

Sterlite was a significant contributor to Vedanta's earnings and profits. Directly or indirectly, the closure of the plant was affecting not only the financial position of the company but also the livelihood of many entities that were dependent on the functioning of the plant. The recurrent protests from the public, and the consequent shutdown of the plant, were also damaging the reputation of the company. What should Vedanta do in such a hostile environment? Should it approach the National Green Tribunal against the state government's order? Or should it relocate its plant to a different state? What strategies should it adopt to improve its image?

Veena Keshav Pailwar is a faculty member at the Institute of Management Technology, Nagpur, India.
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EXHIBIT 1: INDUSTRYWISE CONSUMPTION OF COPPER IN INDIA IN 2015–2016 (%)

Source: Government of India, Ministry of Mines, Indian Bureau of Mines, *Indian Minerals Yearbook 2017 (Part-II: Metals & Alloys)*, 56th ed., Copper (Final Release), March 2018, accessed February 16, 2019, <http://ibm.nic.in/writereaddata/files/01092019101204Copper%202017%20.pdf>.

EXHIBIT 2: STATES AND UNION TERRITORIES* IN INDIA

State	Population	Per Capita Income (Rank)	Ease of Doing Business Rank	Coastal Length (in km)/Main Port**
Andhra Pradesh	84,580,777	95,566 (11)	1	973.7/Visakhapatnam
Arunachal Pradesh	1,383,727	86,818 (25)	34	-
Assam	31,205,576	52,416 (18)	17	-
Bihar	104,099,452	28,485 (21)	18	-
Chhattisgarh	25,545,198	71,581 (13)	6	-
Goa	1,458,545	308,827 (22)	19	181.48/Marmagao and Panji
Gujarat	60,439,692	131,853 (4)	5	2,125.87/Kandla
Haryana	25,351,462	143,211 (1)	3	-
Himachal Pradesh	6,864,602	119,386 (8)	16	-
Jammu & Kashmir	12,541,302	59,924 (15)	22	-
Jharkhand	32,988,134	49,174 (19)	4	-
Karnataka	61,095,297	120,496 (7)	8	309.59/Panambur
Kerala	33,406,061	128,550 (5)	21	597.43/Cochin
Madhya Pradesh	72,626,809	53047 (17)	7	-
Maharashtra	112,374,333	133,141 (3)	13	896.98/Jawaharlal Nehru Port
Manipur	2,570,390	46,756 (28)	32	-
Meghalaya	2,966,889	58,826 (27)	34	-
Mizoram	1,097,206	101,877 (24)	30	-
Nagaland	1,978,502	63,568 (26)	28	-
Odisha	41,974,218	63,674 (14)	14	667.12/Pradip
Punjab	27,743,338	105,386 (10)	20	-
Rajasthan	68,548,437	72,072 (12)	9	-
Sikkim	610,577	206,178 (23)	33	-
Tamil Nadu	72,147,030	117,806 (9)	15	1,069.98/Tuticorin, Chennai, and Ennore
Telangana	N/A	122,684 (6)	2	-
Tripura	3,673,917	N/A (29)	25	-
Uttar Pradesh	199,812,341	38,934 (20)	12	-
Uttarakhand	10,086,292	133,246 (2)	11	-
West Bengal	91,276,115	61,245 (16)	10	662.90/Haldia
Union Territories				
Andaman & Nicobar Islands	3,80,581	113,796 (33)	31	3,078.52/ Port Blair
Chandigarh	1,055,450	207,000 (31)	29	
Daman and Diu		N/A	24	63.10
Dadra & Nagar Haveli		N/A	26	
Delhi	16,787,941	249,555 (30)	23	
Lakshadweep		N/A	34	121.92
Punducherry	1,247,953	137,088 (32)	27	42.11
All India	1,210,569,573	82,229		11,084

Note: *India is a Union of States. The president is the constitutional head of the Executive of the Union, whereas the governor is the head of the Executive of the State. The states are ruled by the state governments while the Union Territories are administered by the president. **It can be noted that the increasing level of organic and inorganic pollution in the coastal water over the years is a major concern for most of the states with major ports in the country.

Source: Compiled from "State Wise Data," updated February 2, 2020, accessed January 30, 2020, www.esopb.gov.in/Static/PDF/GSDP/Statewise-Data/StateWiseData.pdf; HT Correspondent, "Andhra Pradesh Tops Ease of Doing Business Rankings for States, Telangana Second," *Hindustan Times*, July 10, 2018, accessed February 15, 2020, <https://www.hindustantimes.com/business-news/andhra-pradesh-tops-state-wise-list-of-ease-of-doing-business-in-india/story-Erg2rVXwMhaylywHeUQ8IJ.html>; Central Water Commission, *Status Report on Coastal Protection and Development in India*, December 2016, accessed February 16, 2020, http://old.cwc.gov.in/CPDAC-Website/Paper_Research_Work/Status_Report_on%20Coastal_Protection_and%20Development_in%20India_2016.pdf; "Sea Ports in India," Mapsofindia, accessed February 16, 2020, <https://www.mapsofindia.com/maps/sea-ports/>.

EXHIBIT 3: PRODUCERS OF COPPER METAL, 2016–2017

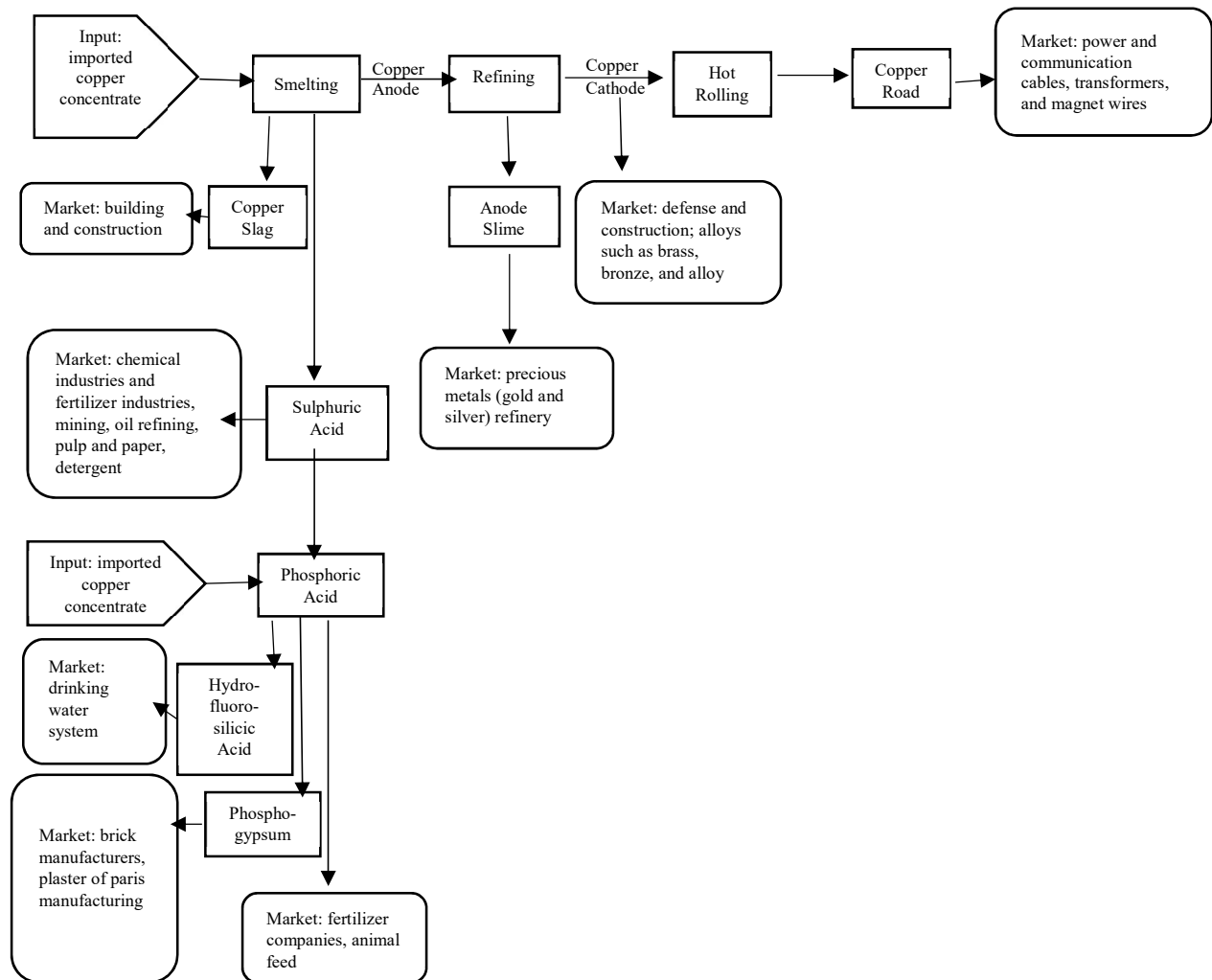
Producer	Location	
	State	District
Hindustan Copper Limited	Jharkhand Maharashtra	Singhbhum (East) Raigad
Hindalco Industries Limited	Gujarat	Bharuch
Vedant(Sterlite)	Tamil Nadu Dadra & Nagar Haveli	Chinchpada (Silvassa)Thoothukudi

Source: Government of India, Ministry of Mines, Indian Bureau of Mines, *Indian Minerals Yearbook 2017 (Part-II: Metals & Alloys)*, 56th ed., Copper (Final Release), March 2018, accessed February 16, 2019, <http://ibm.nic.in/writereaddata/files/01092019101204Copper%202017%20.pdf>.

EXHIBIT 4: ANNUAL CAPACITY OF COPPER SMELTERS (IN '000 TONNES)

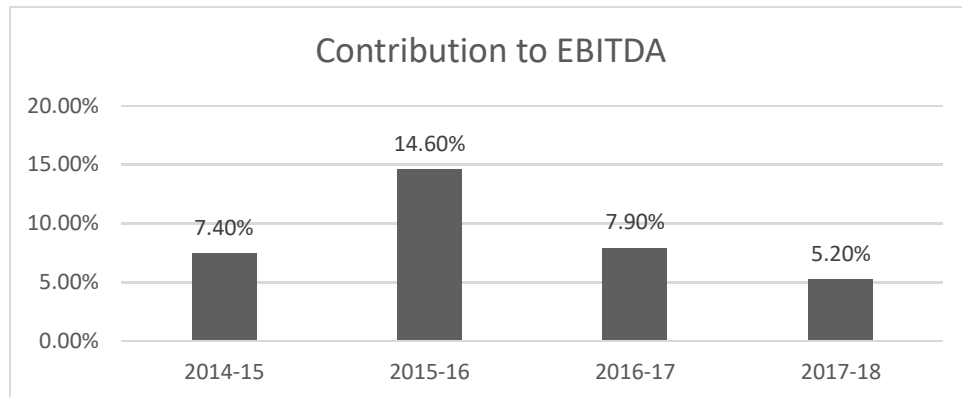
Company	Annual Capacity
Total	1,001.5
Hindustan Copper Limited	101.5
Hindalco Industries Limited	500.0
Sterlite(or Vedanta)	400.0

Source: Government of India, Ministry of Mines, Indian Bureau of Mines, *Indian Minerals Yearbook 2017 (Part-II: Metals & Alloys)*, 56th ed., Copper (Final Release), March 2018, accessed February 16, 2019, <http://ibm.nic.in/writereaddata/files/01092019101204Copper%202017%20.pdf>

EXHIBIT 5: VALUE CHAIN OF STERLITE COPPER SMELTER

Source: Based on information available at "Product," Sterlite Copper accessed August 3, 2020, <https://www.sterlitecopper.com/our-business/product/by-product/>.

"Products," Vedanta: Sterlite Copper, accessed August 3, 2020, <https://www.sterlitecopper.com/our-business/products/>.

EXHIBIT 6: STERLITE COPPER PLANT—CONTRIBUTION TO VEDANTA'S EBITDA

Note: EBITDA = earnings before interest, tax, depreciation, and amortization.

Source: R. Sree Ram, "NGT Lifeline for Vedanta Copper Business Awaits Political Test," Mint, December 19, 2018, accessed February 16, 2020, <https://www.livemint.com/Money/tag76BwHopQ6SfQpBneQxJ/NGT-lifeline-for-Vedantas-Sterlite-copper-plant-in-Thoothuk.html>.

EXHIBIT 7: THE NATIONAL GREEN TRIBUNAL (NGT)

The Government of India established the *National Green Tribunal* (NGT) in 2010 with the following objectives:

- To provide an effective and expeditious resolution for cases pertaining to environmental protection and natural resources.
- To enforce legal rights relating to the environment.
- To provide relief and compensation to the parties affected by environmental damage.

The tribunal, which consisted of a full-time chairperson, judicial members, and expert members, was mandated to complete and dispose of the applications or appeals within six months of their filing. Its authority was equivalent to that of a civil court. Any entity aggrieved by the decision of the tribunal could appeal to the Supreme Court within 90 days of the commencement of the award. Even the high courts had the jurisdiction to entertain appeals against the orders of the tribunal.

The principal bench of the NGT was located in New Delhi, while the regional benches were in Pune (Western Zone Bench), Bhopal (Central Zone Bench), Chennai (Southern Bench), and Kolkata (Eastern Bench).

Source: Compiled from: Praveen Bhargav, "Everything You Need To Know About the National Green Tribunal (NGT)," Conservation India, May 2, 2011, accessed May 2, 2019, www.conservationindia.org/resources/ngt; Rahi Gaikwad; "Challenges to NGT Orders in Courts a Growing Concern," *The Hindu*, March 23, 2016, accessed May 2, 2020, <https://www.thehindu.com/news/cities/mumbai/news/challenges-to-ngt-orders-in-courts-a-growing-concern/article8387736.ece>.

EXHIBIT 8: SHAREHOLDING PATTERN OF VEDANTA

Category of Shareholder	Total No. of Shareholders
Shareholding of Promoter and Promoter Group	8
Indian	4
Foreign	4
Public Shareholding	
Institutions	690
Mutual Funds	38
Financial Institutions/Banks	48
Insurance Companies	4
Qualified Foreign Investors	596
Others	4
Non-institutions	529,360
Individuals	507,410
Others	21,950

Source: "Vedanta Ltd.," Moneycontrol, accessed January 28, 2019, <https://www.moneycontrol.com/company-facts/vedanta/shareholding-pattern/SG>.

ENDNOTES

- ¹ This case has been written on the basis of published sources only. Consequently, the interpretation and perspectives presented in this case are not necessarily those of Sterlite Copper or any of its employees.
- ² Bharani Vaitheesvaran and Rakhi Mazumdar, "Tuticorin Protest: Tamil Nadu Government Orders Permanent Closure of Sterlite Plant," *The Economic Times*, May 29, 2018, accessed January 28, 2020, <https://economictimes.indiatimes.com/news/politics-and-nation/tamil-nadu-government-orders-permanent-closure-of-sterlite-plant-in-tuticorin/articleshow/64355730.cms>.
- ³ Ibid.
- ⁴ ₹ = INR = Indian rupee; US\$1 = ₹67.445 on May 31, 2018.
- ⁵ Gireesh Babu, "Sterlite Copper's Closure to Impact Turnover of Rs 600 Bn for Industry: CEO," *Business Standard*, July 23, 2018, accessed August 1, 2020, https://www.business-standard.com/article/companies/sterlite-copper-s-closure-to-impact-turnover-of-rs-600-bn-to-industry-ceo-118072200153_1.html.
- ⁶ Ibid.
- ⁷ "Sterlite Protest in Thoothukudi: False Propaganda Dominated over Truth about the Company: Says CEO P. Ramnath," *First Post*, May 31, 2018, accessed January 28, 2020, <https://www.firstpost.com/business/sterlite-protest-in-thoothukudi-false-propaganda-dominated-the-truth-about-the-company-says-ceo-p-ramnath-4489863.html>.
- ⁸ "Birla Copper Sees Indian Copper Demand Doubling by 2026," *Reuters*, November 14, 2018, accessed October 20, 2020, <https://in.reuters.com/article/china-copper-birla-copper/birla-copper-sees-indian-copper-demand-doubling-by-2026-idUSL4N1XQ1DS>.
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